

DMB PRODTRACK

Business Document

Purpose, production context, processes, rules, KPIs and scope

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Audience	Plant management, planners, supervisors, quality, IT and project stakeholders
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Practice project by DMB Websolutions. The manufacturing scenario, business rules and data are fictitious, plausible assumptions.

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Executive summary

DMB ProdTrack is a production work-order and shop-floor tracking system for a plant that makes safety identification products - pipe markers, valve tags, safety signs and labels. It replaces paper travelers and spreadsheets with a single system that follows each job from the customer's sales order through artwork approval, release, station-by-station execution, quality inspection and completion, and shows the live state of the floor on a dashboard.

This document describes the application **as built** after the Sprint 2-3 batch (27 September 2026). It is a practice project by DMB Websolutions: the plant scenario, business rules and data are fictitious, plausible assumptions for a marking-products plant.

What the current release delivers:

- Sales orders with lines, and one-click generation of work orders from the lines.
- Work orders with specification snapshot, artwork proof approval, release to the floor, hold/resume/cancel with reason codes, and a printable traveler with QR codes.
- A shop-floor web app for tablets: station queue, QR scanning (camera, hardware scanner or typed), start, pause, complete with good quantity, scrap with reason codes, and QC checklists.
- A live supervisor dashboard (work orders by status, late jobs, today's good/scrap units and scrap rate, WIP by station) that updates as operators work.
- User administration with six roles, a full audit trail, and protection against two people overwriting each other's changes.

Purpose and problem statement

A marking-products plant handles many small, highly customised orders. One customer order can contain hundreds of pipe-marker legends, engraved valve tags with unique numbers and safety signs in several sizes and materials, often for oil and gas, power and industrial projects with strict deadlines and documentation requirements.

In the assumed current situation:

- Orders are re-keyed from sales documents into spreadsheets and work instructions travel on paper.
- Supervisors walk the floor to find out where a job is; work in progress per station is unknown.
- Late jobs are discovered late, and customer service cannot quickly answer "where is my order?".
- Scrap is recorded on paper, if at all, so its main causes are not measured.
- QC results are kept on paper forms, which makes traceability slow.

Purpose of ProdTrack: provide real-time, reliable and auditable information about production status and quality, with minimal typing on the shop floor.

Business objectives

ID	Objective	How the current release supports it
BO1	Real-time visibility of every work order	Operation events are pushed to the dashboard and work order pages within seconds (SignalR)
BO2	Improve on-time delivery	Late work orders are counted on the dashboard and highlighted in lists; at-risk forecasting is future work
BO3	Reduce scrap through measurement	Every scrap entry needs a quantity and a reason code; scrap units and scrap rate are shown for today
BO4	Paperless QC traceability	QC checklists are recorded per operation with inspector, time, measured values and result
BO5	Reduce data entry	Operators scan the traveler QR codes instead of typing numbers

Production context

Product families

Family	Examples	Typical specification
Pipe markers	Printed vinyl markers for flammable, water, compressed air lines	Material, pipe OD range, ASME A13.1 colour scheme, legend text, letter height
Valve tags	Engraved brass or aluminium tags	Material, shape, diameter, thickness, hole size, engraved text
Safety signs	Aluminium or polymer signs	Size, material, ANSI Z535 signal word (DANGER, WARNING, CAUTION, NOTICE), mounting
Labels	Polyester labels	Size, adhesive, finish

Reference data seeded with the application includes the ASME A13.1 colour schemes (for example Flammable = black on yellow, Fire quenching = white on red) and the ANSI Z535 signal words with their colours.

Stations and routings

Station	Type	Used by
PREPRESS	Prepress / artwork	All product types
PRINT-01	Digital printing	Pipe markers, signs, labels
LAM-01	Laminating	Pipe markers, signs
ENGRAVE-01	Engraving	Valve tags
DIECUT-01	Die-cutting / cutting	Pipe markers, signs, labels
QC-01	Quality control (inspection)	All product types
PACK-01	Packing and shipping	All product types

A **routing** is the ordered list of stations for a product type, with setup time and standard minutes per unit. For example, a printed pipe marker goes PREPRESS, PRINT-01, LAM-01, DIECUT-01, QC-01, PACK-01; an engraved valve tag goes PREPRESS, ENGRAVE-01, QC-01, PACK-01. Routings are versioned: planners edit them by creating a new version.

Stakeholders and roles

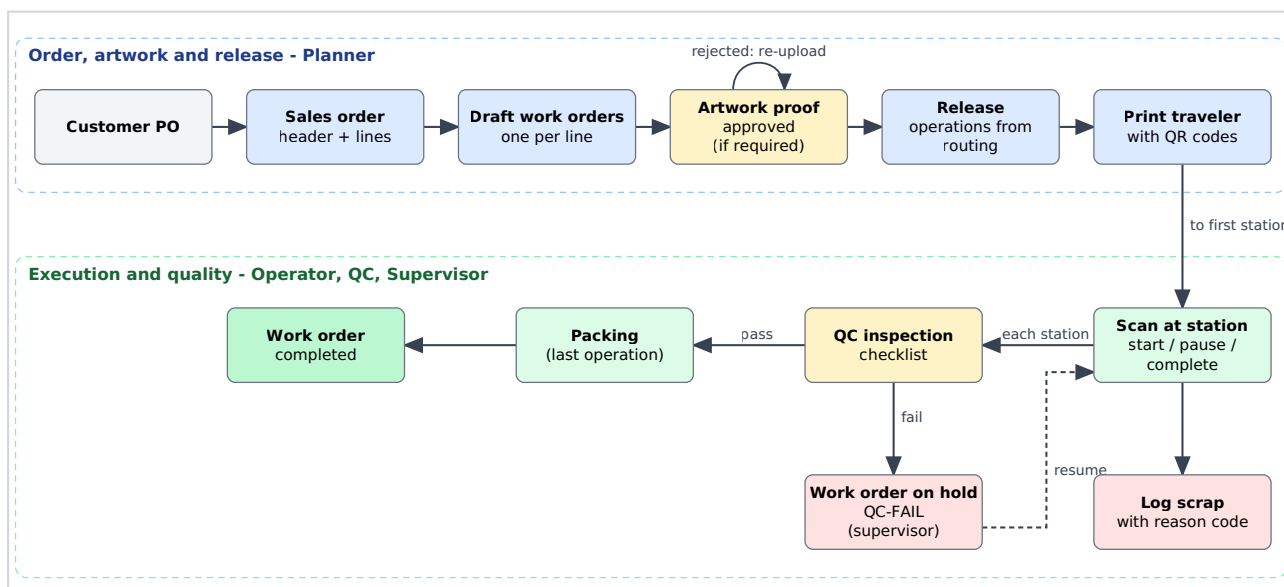
Stakeholder	Interest	Role in ProdTrack
Plant manager	Throughput, on-time delivery, cost	Viewer or Supervisor (dashboard)
Production planner	Order entry, artwork, releasing work	Planner
Shift supervisor	WIP, bottlenecks, holds	Supervisor
Machine operator	Clear queue, easy recording	Operator (shop-floor app)
QC inspector	Consistent inspections, stopping bad product	QC
Customer service / sales	Order status for customers	Viewer
IT administrator	Users, security, availability	Admin
Customers (indirect)	On-time, correct, compliant product	-

Role permissions

Capability	Admin	Planner	Supervisor	Operator	QC	Viewer
View dashboard, orders, products, routings	Yes	Yes	Yes	Yes	Yes	Yes
Sales orders and work orders (create, edit, release)	Yes	Yes	-	-	-	-
Products and routings (edit)	Yes	Yes	-	-	-	-
Hold / resume / cancel work orders, print traveler	Yes	Yes	Yes	-	-	-
Upload artwork proof	Yes	Yes	-	Yes	-	-
Approve / reject artwork	Yes	Yes	-	-	-	-
Run operations (start, pause, complete, scrap)	Yes	-	Yes	Yes	Yes	-
Record QC inspections	Yes	-	-	-	Yes	-
Stations, reason codes, users	Yes	-	-	-	-	-

Business processes

End-to-end flow



Sales order to completed work order

1. Sales order entry

The planner enters the customer's purchase order as a **sales order** (S0-yyyy-nnnnn): customer, customer PO, due date and one line per product with quantity and legend text. The product's default specification is copied onto the line and can be adjusted.

2. Work order generation

Create work orders turns each sales order line into a Draft **work order** (W0-yyyy-nnnnnn) with the line's product, quantity, legend, specification and due date. The specification is a snapshot: later catalogue changes do not change existing work orders. Lines that already have a work order are skipped. Planners can also create work orders directly from a product (stock or internal jobs).

3. Artwork approval

For products flagged requires artwork approval, the planner or operator uploads a proof (PDF or image). A planner approves it or rejects it with a reason. Only an approved proof allows release. A new upload supersedes the previous proof.

4. Release and traveler

Release copies the current routing into the work order as numbered operations. The first operation becomes Ready at its station. The planner prints the **traveler**, which carries a QR code for the work order and one per operation, the specification and space for handwritten notes.

5. Operations at stations

At each station the operator opens the queue on the tablet, scans the traveler and:

- **starts** the operation (the operator and start time are recorded);
- **pauses** it with a reason when needed, and resumes;
- **logs scrap** with a quantity and a scrap reason;
- **completes** it with the good quantity. The next operation becomes Ready with that quantity as its input.

6. Quality control and scrap

At the QC station the inspector fills in the checklist for the product type. A pass completes the inspection step and the job moves to packing. A fail puts the whole work order **on hold** with reason **QC-FAIL** so a supervisor can decide what to do. Scrap is captured at every station with reason codes so its causes can be analysed.

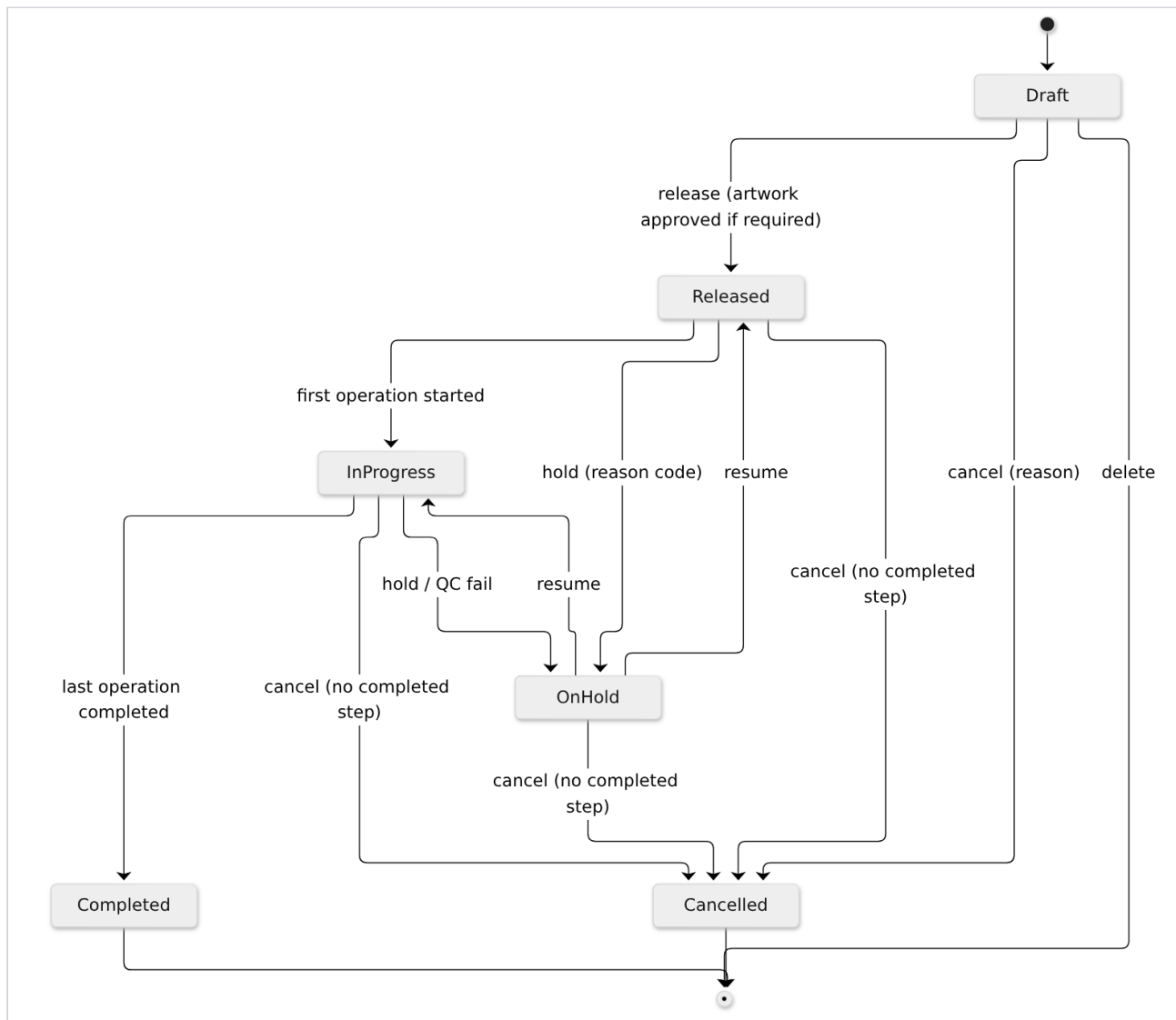
7. Hold, resume and cancel

Supervisors and planners can put a work order on hold with a hold reason (for example a customer change), resume it (back to its previous status) or cancel it with a reason while no step has been completed.

8. Completion

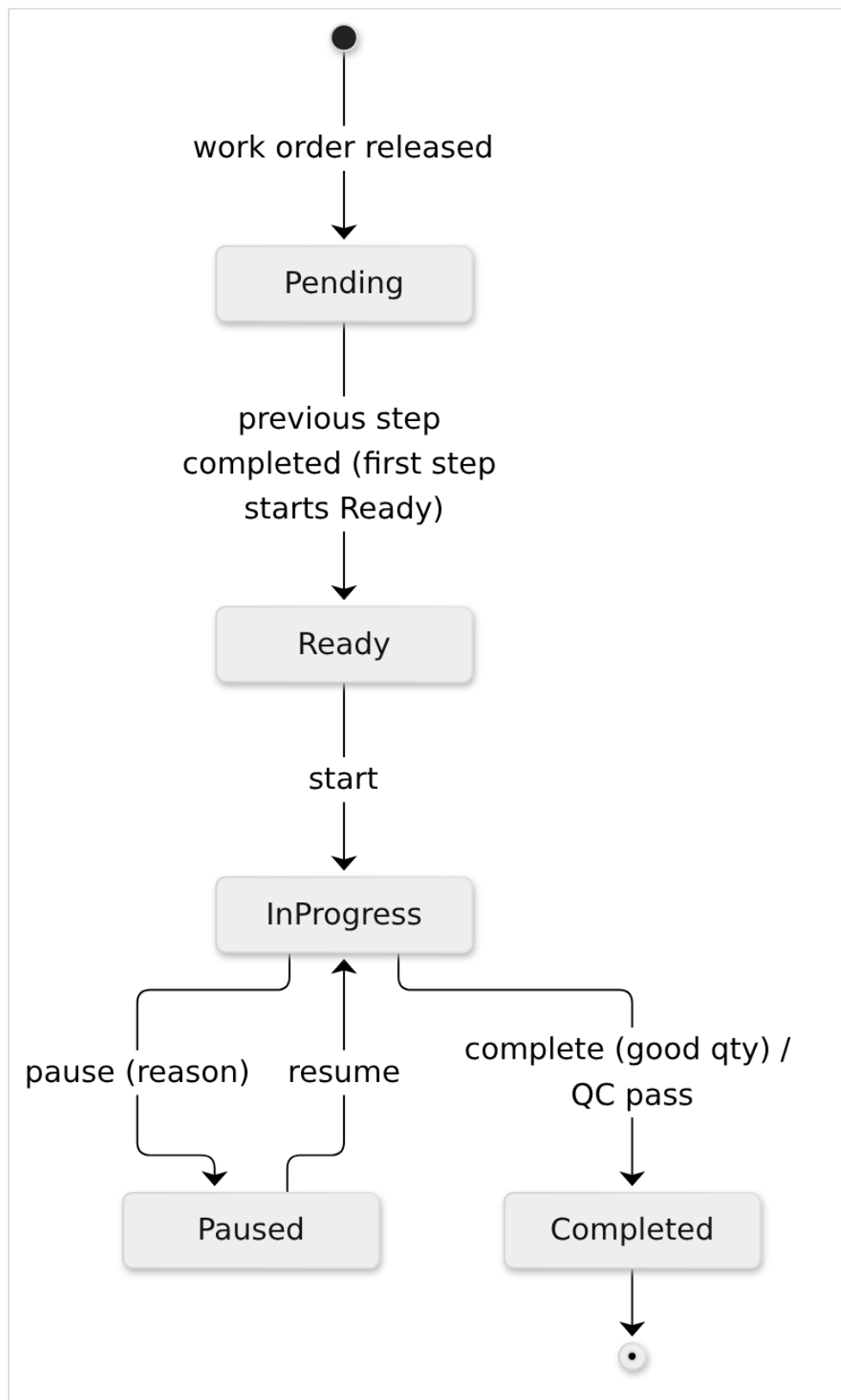
Completing the last operation (packing) completes the work order; its completed quantity is the good quantity of that last step. The sales order shows each line's work order status.

Work order lifecycle



Work order status transitions

Operation lifecycle



Operation status transitions

Business rules

ID	Rule (as implemented)
BR-01	Numbers: work order W0-{yyyy}-{nnnnnn} , sales order S0-{yyyy}-{nnnnn} , generated by the server and safe under concurrent use.
BR-02	The product specification is snapshotted onto the sales order line and the work order; catalogue changes do not alter existing orders.
BR-03	An operation can only be started when it is Ready and its work order is Released or In progress (not On hold, Cancelled or Completed). If a station is given, it must be the routed station.
BR-04	The first operation is Ready at release; each next operation becomes Ready when the previous one completes.
BR-05	The input quantity of an operation is the good quantity of the previous one (first = work order quantity). Good plus scrap cannot exceed the input.
BR-06	A work order is Completed when its last operation completes; its completed quantity is the last operation's good quantity.
BR-07	Server UTC time is authoritative for all events; dates are displayed in plant time (Asia/Manila).
BR-08	Pauses and scrap require an active reason code of the matching category (Pause, Scrap). Holds require a Hold reason code; cancel requires a free-text reason.
BR-09	QC measured values outside [min, max] fail the item automatically; any failed item fails the inspection.
BR-10	A failed inspection puts the work order on hold with reason QC-FAIL . An inspection step can only be completed after a passed inspection.
BR-11	Products flagged "requires artwork approval" cannot be released without an approved proof; approving a new version supersedes the old one.
BR-12	Cancelling a work order is allowed only while no operation is completed. Cancelling a sales order is allowed only while it has no open work orders.
BR-13	Sales order lines that already have a work order cannot be changed or removed.
BR-14	Routing changes create a new version; released work orders keep the operations they were released with.
BR-15	Users are created by an administrator only; there is no self-registration and no email. New and reset passwords must be changed at the next sign-in.
BR-16	Every change to business data is recorded in the audit trail (who, when, what changed, correlation id).
BR-17	When two users edit the same record, the second save is rejected with a "changed by someone else" message instead of silently overwriting the first.

Key performance indicators

KPI	Definition	Where shown
Work orders by status	Count of work orders in Draft, Released, In progress, On hold, Completed	Dashboard tiles
Late work orders	Open (not Completed/Cancelled) with due date before today (plant time)	Dashboard tile; lists highlight late rows
Completed today	Work orders completed since midnight plant time	Dashboard
Good units today	Sum of good quantity of operations completed today	Dashboard
Scrap units today	Sum of scrap quantity logged today	Dashboard
Scrap rate today	scrap / (good + scrap) for today, e.g. $2 / (114 + 2) = 1.7 \%$	Dashboard
WIP by station	Operations Ready, In progress, Paused and Pending per station	Dashboard table

Planned KPIs for later releases: throughput per period and station, on-time delivery, at-risk jobs, scrap Pareto by reason, OEE-lite (availability and performance from downtime and standard minutes).

Scope

In scope (delivered)

Area	Delivered capability
Security	Sign-in with personal accounts, forced password change, lockout, six roles, administrator-managed users
Master data	Products with type-specific specifications, stations, reason codes, ASME/ANSI reference data, versioned routings with editor, QC checklist templates (seeded)
Orders	Sales orders with lines, work order generation, work orders with specification snapshot
Artwork	Proof upload, approval, rejection, versioning
Release	Operations from routing, printable traveler with QR codes
Shop floor	Installable tablet app, station queue, scanning, start/pause/resume/complete, scrap, QC inspections
Control	Hold/resume/cancel with reason codes, automatic hold on failed QC
Visibility	Live dashboard and work order pages
Compliance	Audit trail of all changes, optimistic concurrency

Out of scope for this release / future

Item	Notes
Materials and inventory	Stock levels, receipts, consumption and low-stock alerts (backlog PT-038 to PT-040, PT-049)
Downtime and OEE-lite	Station downtime logging and OEE-lite KPIs (PT-045, PT-046)
Rework routing	Failed QC currently holds the job; creating rework operations is future work
Reporting	Late/at-risk list, throughput and scrap Pareto charts, work order timeline, audit viewer (PT-042 to PT-044, PT-048)
Traveler reprint audit	Reprints are possible from the browser but not yet logged (PT-026)
Badge/PIN sign-in on tablets	Tablets use the normal account sign-in today (PT-011)
Email notifications	Not available on the free hosting plan; administrators hand out passwords
ERP / accounting integration	Not planned for the MVP
Hosting	The application currently runs locally. Deployment to the MonsterASP.NET free plan and an Azure DevOps phase 2 are documented options, not yet performed

Assumptions and constraints

- Single plant, single time zone (Asia/Manila), one shift pattern.
- Shop-floor devices are tablets or phones with a modern browser (Chrome or Edge recommended for camera scanning).
- Cost must stay at zero: free, open-source libraries (MIT or Apache licensed) and free hosting tiers only.
- No email or SMS is available; all notifications are in-app.
- All data in this practice project is fictitious.

Glossary

Term	Meaning
Sales order (SO)	A customer's order, with one line per product
Work order (WO)	A production job for one product and quantity
Operation	One step of a work order at a station (for example printing)
Routing	The ordered stations and standard times for a product type
Traveler	The printed sheet that travels with a job, carrying QR codes
WIP	Work in progress
Scrap	Units that are unusable, recorded with a reason code
Hold	A temporary stop of a work order with a reason
QC	Quality control inspection
ASME A13.1	US standard for pipe marker colours and sizes
ANSI Z535	US standard for safety sign signal words and colours